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Reduced stature maize and mads-box transcription factors

Abstract

The compositions and methods relate to shorter stature corn plants having an increased and extended expression of a MADS-box polypeptide along with one or more genetic modifications that result in shorter stature.

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Primary Examiner: Collins; Cynthia E

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of PCT Application Serial Number PCT/US2020/064999, filed Dec. 15, 2020, which claims the benefit of U.S. Provisional Application No. 62/949,206, filed Dec. 17, 2019, both of which are hereby incorporated herein by reference in their entireties.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY
(1) The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII formatted sequence listing with a file named “8197_ST25.txt” created on Dec. 17, 2019 and having a size of 39 kilobytes and is filed concurrently with the specification. The sequence listing contained in this ASCII formatted document is part of the specification and is herein incorporated by reference in its entirety.
FIELD
(2) Embodiments disclosed herein relate to the field of plant molecular biology and agronomic traits.
BACKGROUND
(3) Corn is an agriculturally important crop and serves as a food and feed source for animal, human, and industrial uses. Increased grain yield may be achieved in maize plants by a variety of ways, including expression of a transgene to increase grain yield in addition to improved breeding. Performance of a transgene in a plant including the agronomic parameters, may be impacted by a variety of factors such as the use of expression elements including promoter/regulatory elements, the genomic location of the insert sequence, copy number of the inserted transgene and genetic (germplasm) and environmental factors such as soil, temperature, light and moisture. The identification of constructs, testing of orthologs and transformation events that result in increased grain yield of a maize plant at a commercially relevant level in the field are the result of a substantial and significant developmental effort towards product advancement. Accordingly, it would be desirable to have maize plants that demonstrate increased grain yield.
SUMMARY
(4) A corn plant comprising a stature reducing genetic modification and a yield enhancing genetic modification, wherein the yield enhancing genetic modification comprises a heterologous regulatory polynucleotide operably linked to a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 95% identical to SEQ ID NO: 1 and wherein the corn plant exhibits a plant height reduction of about 5% to about 50%, when compared to a control corn plant not comprising the stature reducing genetic modification. In certain embodiments, the stature reducing genetic modification is an introduced genetic modification in one or more genomic loci selected from the group consisting of: a polynucleotide encoding the D8 polypeptide; a polynucleotide encoding a MYB transcription factor expression or activity; MYB transcription factor genomic locus comprises a brachytic 1 (br1) allele in maize; a component involved in a

biological pathway related to gibberellic acid biosynthesis or gibberellic acid signaling; a component involved in a biological pathway related to auxin transport, auxin signaling; a component involved in a biological pathway related to brassinosteroid biosynthesis or signaling; and Brachytic 2 (Br2) allele.

(5) A seed produced from the corn plant wherein a progeny from the seed exhibits increased yield and reduced stature when compared to the control plant. A corn plant, seed, cell or part thereof comprising event DP-202216-6 and exhibiting reduced stature, wherein a representative sample of seed of said corn event has been deposited with American Type Culture Collection (ATCC) with Accession No. PTA-124653.

(6) A unit of corn seeds, the unit comprising a mixture of seeds wherein the mixture comprises about 10% to about 75% seeds exhibiting reduced stature due to an introduced genetic modification and exhibiting increased yield due to increased expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 95% identical to SEQ ID NO: 1, when compared to a control plant.

(7) A method of increasing planting density of corn plants comprising a heterologous regulatory polynucleotide operably linked to a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 95% identical to SEQ ID NO: 1, the method comprising:

(8) a) providing corn plants wherein the expression of the polynucleotide encoding the polypeptide is increased when compared to a control plant;

(9) b) reducing plant height by introducing a genetic modification that results in reduced stature of the corn plants; and

(10) c) planting the corn plants at a planting density of about 30,000 to about 75,000 plants per acre.

(11) In certain embodiments, the planting density is at least 50,000 plants; 55,000 plants; 58,000 plants; 60,000 plants; 62,000 plants; 64,000 plants. In certain aspects, the corn plants comprise a mutation in a genomic region encoding D8 polypeptide or reduced expression of the polynucleotide encoding D8 polypeptide. In certain aspects, the corn plants are planted in a plurality of rows having a row width of about 8 inches to about 30 inches. In certain aspects, the corn plants yield an average of about 3 bu/acre compared to control corn plants.

(12) A method of increasing nitrogen use efficiency of a population of corn plants per unit of applied nitrogen, the method comprising: providing the population of corn plants, wherein the expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 95% identical to SEQ ID NO: 1 is increased due to a genetic modification compared to a control plant; modifying the plant height of the population of the corn plants by introducing a genetic modification that results in reduced stature of the population of the corn plants; and growing the population of plants in a crop growing environment where the applied nitrogen is about 10% to about 50% less than an application rate of about 100 lbs to about 400 lbs of nitrogen per acre. In certain aspects, the population of corn plants exhibit an increase in nitrogen use efficiency of at least about 5% when compared to the control plant. In certain aspects, the nitrogen use efficiency is increased by applying late season nitrogen, wherein at least about 25% of total nitrogen applied is applied on or after V8-V12 stage.

(13) A method of improving an agronomic characteristic of a population of corn plants grown in a field, the method comprising: providing the population of corn plants, wherein the expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 95% identical to SEQ ID NO: 1 is increased due to a genetic modification compared to a control plant; modifying the plant height of the population of the corn plants by introducing a genetic modification that results in reduced stature of the population of the corn plants; and growing the population of plants in a crop growing environment, wherein an agronomic characteristic of the population of the corn plants is improved compared to a control population of plants. In certain aspects, the corn plants exhibit an increase in standability when compared to the control plant. In

certain aspects, the population of corn plants exhibit an increase in early season canopy thereby resulting in improved weed control and/or less applied herbicide per acre. In certain aspects, the population of corn plants exhibit a decrease in early season disease. In certain aspects, the population of corn plants leave reduced net residue per bushel of grain yield after harvest.

(14) A method for reducing production costs of corn hybrids, the method comprising sexually crossing a first parent corn plant with a second parent corn plant, wherein said first parent corn plant comprises event DP-202216-6 DNA and exhibits reduced plant height compared to the second parent corn, thereby producing a plurality of hybrid progeny plants in a field. In certain aspects, the first corn plant is female inbred. In certain aspects, the hybrid yield is increased by at least 5% compared to a control plant.

(15) A reduced stature corn seed includes Event DP-202216-6, wherein a representative sample of corn event DP-202216-6 seed of has been deposited with American Type Culture Collection (ATCC) with Accession No. PTA-124653.

(16) In an embodiment, a corn plant comprises increased and extended expression of a monocot MADS box polynucleotide that increases yield and one or more genetic modifications targeting one or more distinct genomic loci that are involved in plant height reduction. In an embodiment, the plant height is reduced by about 5% to about 30% compared to the control plant. In an embodiment, such plant comprises an average leaf length to width ratio reduced at V6-V8 growth stages. In an embodiment, such plant height reduction does not substantially affect flowering time. In an embodiment, the flowering time does not change by more than about 5-10 CRM or plus or minus 10% GDU or 125-250 GDU, compared to a control plant not comprising the modifications or the increased MADS polypeptide expression, wherein 25 GDU is equivalent to about 1 day and 1 CRM is about 1 day. In an embodiment, the plant height reduction does not substantially alter root architecture of the plant or does not significantly increase root lodging, compared to a control plant not comprising the modifications. In an embodiment, the plant is substantially tolerant to lodging as measured at a single plant level or at an increased planting density, compared to a control plant or control population of plants. In an embodiment, the plant comprises up to about 10% less number of leaves compared to the control plant. In an embodiment, the plant is maize and the plant height reduction is characterized by the shortening of distance between one or more internodes that are present above or below a female reproductive part of the maize plant. Modified maize plants, whose average internode lengths are reduced compared to the wild-type plants are provided. For example, average internode length (2nd internode length and/or 4th internode length relative to the position of the ear) that is at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75% less than the same or average internode length of a wild-type or control plant are provided. The 2nd internode” refers to the second internode below the ear of the corn plant, likewise, the “4th internode” refers to the fourth internode below the ear of the corn plant.

(17) Corn plants with increased expression of SEQ ID NO: 1 are provided that have (i) a plant height that is at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75% less than the height of a wild-type or control plant, and/or (ii) a stem or stalk diameter that is at least 5%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80% greater than the stem diameter of the wild-type or control plant.

(18) In an embodiment, in addition to an increase in field grain yield due to extended expression of AG099, one or more of the following agronomic characteristics of the plant is also increased or reduced: harvest index of the plant is increased; leaf area is increased; leaf number above the ear is reduced; ratio of the plant ear height over the plant height is increased; and the yield is increased at higher planting density, as compared to a control plant not comprising the mutations.

(19) In an embodiment, the ear height as measured to the maize plant height is substantially similar or slightly reduced to the ear height measured relative to the control plant height, wherein the plant comprises one or more genetic modifications that increase expression of an endogenous

polynucleotide encoding a polypeptide that is at least 95% identical to SEQ ID NO: 1. In an embodiment, the height reducing genetic modifications target the genomic locus such that more than one genetic modifications are present within (a) the same coding region; (b) non-coding region; (c) regulatory sequence; or (d) untranslated region, of an endogenous polynucleotide encoding a polypeptide that is involved in plant height.
(20) A reduced stature maize plant stably transformed with a recombinant polynucleotide sequence encoding a polypeptide comprising an amino sequence that is at least 90%, 93% 95%, 97%, 98% or 99% identical to SEQ ID NO: 1, wherein the maize plant exhibits increased grain yield compared to a control maize plant not containing the recombinant polynucleotide. In some embodiments, the recombinant polynucleotide is operably linked to a weak heterologous constitutive regulatory element. In some embodiments, the grain yield is at least about three bushels/acre when compared to the control maize plant, wherein the maize plant and the control maize plant are grown in a field under normal crop growing conditions. In some embodiments, the grain yield in the field range from about 2 to about 8 bu/acre when compared to the control population of maize plants grown in a population density of about 20,000 to about 60,000 plants per acre. In some embodiments, the weak heterologous constitutive regulatory element is a maize GOS2 promoter. In some embodiments, the amino acid sequence is at least 95% identical to SEQ ID NO: 1 and the maize plant comprises a polynucleotide encoding a polypeptide that provides herbicide tolerance and a polynucleotide that encodes a polypeptide or an RNA sequence that provides resistance to one or more insect pests. Maize seed produced from the maize plant described herein exhibit yield improvement characteristics and reduced stature. In an embodiment, the regulatory element comprises a heterologous intron element.
(21) A method of increasing grain yield of a short stature maize plant, the method comprising expressing a polynucleotide sequence encoding a polypeptide that is at least 90%, 93% 95%, 97%, 98% or 99% identical to SEQ ID NO: 1, wherein the polynucleotide is operably linked to a heterologous regulatory sequence; and growing the maize plant in a field to increase grain yield compared to a control maize plant not containing the polynucleotide operably linked to the heterologous regulatory sequence.
(22) A method of producing a seed, the method comprising the following:
(23) (a) crossing a first plant with a second plant, wherein at least one of the first plant and the second plant comprises a recombinant DNA construct, wherein the recombinant DNA construct comprises (i) a polynucleotide operably linked to at least one regulatory element, wherein the polynucleotide encodes a MADS protein having an amino acid sequence of at least 90% sequence identity, based on the Clustal V or Clustal W method of alignment, when compared to SEQ ID NO: 1 or (ii) a genetic component responsive for reduced stature and wherein the second plant is of shorter stature or reduced height; and
(24) (b) selecting a seed of the crossing of step (a), wherein the seed comprises the recombinant DNA construct.
(25) A reduced stature plant grown from the seed produced by the method described herein, wherein the plant exhibits increased yield, when compared to a control plant not comprising the recombinant DNA construct.
(26) In some embodiments, a method of selecting a plant that exhibits increased yield the method comprises:
(27) (a) obtaining a reduced stature plant, wherein the plant comprises in its genome a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory element, wherein said polynucleotide encodes a MADS protein having an amino acid sequence of at least 90% sequence identity, based on the Clustal V or Clustal W method of alignment, when compared to SEQ ID NO: 1 and a genetic element that reduces plant height;
(28) (b) growing the plant in a field under conditions wherein the polynucleotide is expressed; and
(29) (c) selecting the plant of part that exhibits increased yield when compared to a control plant

not comprising the recombinant DNA construct and exhibits reduction in plant height compared to the control plant.
(30) A reduced stature maize plant that exhibits increased expression of an endogenous polynucleotide encoding a polypeptide comprising a sequence that is at least 95% identical to SEQ ID NO: 1, wherein the increased expression is due to a heterologous regulatory element. In some embodiments, the heterologous regulatory element is a plant-derived enhancer element. In some embodiments, the heterologous regulatory element is a weak constitutive promoter element. In some embodiments, the maize plant is an inbred or a hybrid plant.
(31) In some embodiments, the reduced stature maize plant includes a second polypeptide that provides herbicide tolerance and a third polypeptide that provides insect resistance.
(32) A reduced stature corn plant, seed, cell or part thereof includes event DP-202216-6. A reduced stature corn plant, seed, cell or part thereof includes event DP-202216-6, wherein a representative sample of seed of said corn event has been deposited with American Type Culture Collection (ATCC) with Accession No. PTA-124653. In some embodiments, the plant part is selected from the group consisting of pericarp, pollen, ovule, flower, grain, shoot, root, stalk, silk, tassel, ear, and leaf tissue. A biological sample derived from a reduced stature corn event DP-202216-6 plant, tissue, or seed, wherein a representative sample of said corn event DP-202216-6 seed of has been deposited with American Type Culture Collection (ATCC) with Accession No. PTA-124653. In some embodiments, the biological sample comprises plant, tissue, or portions of seed, pericarp of seed of transgenic corn event DP-202216-6. In some embodiments, the biological sample is a DNA sample extracted from the transgenic corn plant event DP-202216-6. In some embodiments, the first inbred corn line is a female parent or the first inbred corn line is a male parent.
(33) A method for producing a reduced stature corn plant that exhibits increased grain yield in a field, the method comprising: sexually crossing a first parent corn plant with a second parent corn plant, wherein said first or second parent corn plant comprises Event DP-202216-6 DNA or reduced stature, thereby producing a plurality of first generation progeny plants; selfing the first generation progeny plant, thereby producing a plurality of second generation progeny plants; and selecting from the second generation progeny plants that comprise the event DP-202216-6, a plant that exhibits increased grain yield in the field compared to a control corn plant not comprising the event DP-202216-6. In some embodiments, the event DP-202216-6 comprises a recombinant DNA construct and wherein the event DP-202216-6 comprises encodes a polypeptide that is at least 99% identical to SEQ ID NO: 1.
(34) A method of producing hybrid corn seeds comprising: a) sexually crossing a first inbred corn line comprising the DNA construct described herein with a second inbred line not comprising the DNA construct, but exhibits reduced stature; and b) harvesting the hybrid seed produced thereby.
(35) In some embodiments, the step of backcrossing includes backcrossing the second generation progeny plant that comprises reduced stature corn event DP-202216-6 to the parent plant that lacks the corn event DP-202216-6, thereby producing a backcross progeny plant that exhibits increased grain yield compared to a control corn plant not comprising the event DP-202216-6.
(36) A method of increasing grain yield of a population of maize plants in a field, the method comprising growing a population of reduced stature maize plants comprising Event DP-202216-6 in a field and thereby increasing grain yield of the population of maize plants compared to a control plant not comprising the Event DP-202216-6. In some embodiments, the population of reduced stature maize plants are grown under abiotic stress. In some embodiments, the abiotic stress is low nitrogen. In some embodiments, when grown under low nitrogen conditions, the population of reduced stature maize plants comprising the Event DP-202216-6 exhibits yield stability compared to the control population of plants grown under low nitrogen. In some embodiments, the low nitrogen is about 25% to about 75% reduction in the amount of nitrogen normally applied to grow hybrid corn plants in the field. In some embodiments, the reduction in nitrogen applied to field ranges from about 5% to about 10%, 20%, 30%, 40%, 50%, 60% or 70% compared to a normal

application of nitrogen.

(37) According to another embodiment, methods of producing a corn plant that comprise the steps of: (a) sexually crossing a first parental corn line comprising the expression cassettes disclosed herein, which increase yield, and a second parental corn line that lacks such constructs but exhibits reduced stature, thereby producing a plurality of progeny plants; and (b) selecting a progeny plant that shows increase in yield. Such methods may optionally include the further step of back-crossing the progeny plant to the second parental corn line to producing a true-breeding corn plant that exhibits yield increase.

(38) In an aspect, a method of growing a population of corn plants in a field, the method comprising providing seeds that grow as corn plants that comprise (a) introduced genetic modification that increases and extends the expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that encodes a MADS-box monocot transcription factor; (b) one or more introduced genetic modifications that reduced plant height by about 10% to about 50% compared to a control corn plant not comprising the height reducing genetic modifications; and (c) root depth that is at least 50% of the root depth of a control plant not comprising the height reducing genetic modifications. In an aspect, the polypeptide comprises an amino acid sequence that is at least 90% identical to SEQ ID NO: 1. In an aspect, the field comprises about 40,000 to about 70,000 corn plants per acre. In an aspect, the field comprises a root growth enhancing agricultural composition. In an aspect, the agricultural composition is a biological applied as a seed treatment or soil applied. In an aspect, the population of plants comprise one or more introduced genetic modification that increases root growth.

(39) In an aspect, a method of managing disease in a field comprising short stature, high yielding corn plants planted at a high planting density, the method comprising (i) providing seeds that grow as corn plants that comprise (a) introduced genetic modification that increases and extends the expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that encodes a MADS-box monocot transcription factor; and (b) one or more introduced genetic modifications that reduced plant height by about 10% to about 50% compared to a control corn plant not comprising the height reducing genetic modifications; and (ii) providing a pesticide composition at an effective rate such that the pesticide is available to control disease occurrence during mid or late growth season in the field, wherein the corm plants are planted at a planting density of at least 45,000 plants per acre. In an aspect, the pesticide is applied as an extended release pesticide that controls one or more corn diseases. In an aspect, the pesticide is applied over-the-top of the corn plants. In an aspect, the pesticide is applied as a seed treatment. In an aspect, the corn plants comprise one or more introduced genetic modifications that provide resistance to one or more diseases that occur within short stature, high-yielding and at high planting density. In an aspect, the corn plants comprise a transgene that provides resistance to one or more pests. In an aspect, the corn plants are subject to an abiotic stress. In an aspect, the corn plants are subject to insect pressure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic illustration showing N-terminal deletion genome edits to create short stature/semi-dwarf corn plants. (A) shows the illustration where a deletion of 579 bp with insertion of flipped 39 bp of 5′UTR & 5 bp CDS. The genomic length after the edit is 8170 bp. (B) shows the genomic region of the wild-type, approximately about 9000 bp in length.

BRIEF DESCRIPTION OF THE SEQUENCES

(2) The disclosure can be more fully understood from the following detailed description and the accompanying drawings and Sequence Listing that form a part of this application.

(3) The sequence descriptions summarize the Sequence Listing attached hereto, which is hereby incorporated by reference. The Sequence Listing contains one letter codes for nucleotide sequence characters and the single and three letter codes for amino acids as defined in the IUPAC-IUB standards described in Nucleic Acids Research 13:3021-3030 (1985) and in the Biochemical Journal 219(2):345-373 (1984).
(4) TABLE-US-00001 TABLE 1 Sequence Listing Description SEQ ID NO: Description 1 Maize MADS-box protein 28 2 Maize MADS-box protein 28 DNA 3 ZM-D8_CDS_B73 4 ZM-D8_GENE_B73 5 ZM-D8_PRO_B73 6 ZM-D8-Peptide-B73 7 ZM-BR1_CDS_B73 8 ZM-BR1_PRO_B73 9 Peptide Transcription regulator HTH (ZmBr1)
DETAILED DESCRIPTION
(5) As used herein the singular forms "a", "and", and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a cell" includes a plurality of such cells and reference to "the protein" includes reference to one or more proteins and equivalents thereof, and so forth. All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs unless clearly indicated otherwise.
(6) The disclosure and content of International Patent Application Publication No. WO2019055141A (application serial number PCT/US2018/044498), as it relates to stature modification, are herein incorporated by reference in its entirety.
(7) Compositions of this disclosure include a representative sample of seeds which was deposited as Patent Deposit No. PTA-124653 and plants, plant cells, and seed derived therefrom. Applicant(s) have made a deposit of at least 2500 seeds of maize event DP-202216-6 (Patent Deposit No. PTA-124653) with the American Type Culture Collection (ATCC), Manassas, Va. 20110-2209 USA, on Jan. 12, 2018 and were accepted on Jan. 18, 2018. These deposits were made under the terms of the Budapest Treaty and will be maintained under the terms of the Budapest Treaty on the International Recognition of the Deposit of Microorganisms for the Purposes of Patent Procedure. The seeds deposited with the ATCC on Jan. 12, 2018 were taken from a representative sample deposit maintained by Pioneer Hi-Bred International, Inc., 7250 NW 62.sup.nd Avenue, Johnston, Iowa 50131-1000. Access to this ATCC deposit will be available during the pendency of the application to the Commissioner of Patents and Trademarks and persons determined by the Commissioner to be entitled thereto upon request, in accordance with applicable laws and regulations. All restrictions upon availability to the public will be irrevocably removed upon granting of the patent. Upon issuance of a patent, this deposit of seed of maize Event DP-202216-6 is intended to meet all the necessary requirements of 37 C.F.R. §§ 1.801-1.809, and will be maintained in the ATCC depository, which is a public depository, for a period of 30 years, or 5 years after the most recent request, or for the enforceable life of the patent, whichever is longer, and will be replaced if it becomes nonviable during that period. Upon allowance of any claims in the application, the Applicant(s) will make this deposit available to the public pursuant to the Budapest Treaty. Unauthorized seed multiplication prohibited. The seed may be regulated under one or more applicable National, State or other local regulations and ordinances imposed by one or more competent governmental agencies.
(8) As used herein, the term "corn" means *Zea mays* or maize and includes all plant varieties that can be bred with corn, including wild maize species.
(9) As used herein, the terms "insect resistant" and "impacting insect pests" refers to effecting changes in insect feeding, growth, and/or behavior at any stage of development, including but not limited to: killing the insect; retarding growth; reducing reproductive capability; inhibiting feeding; and the like.
(10) As used herein, the terms "pesticidal activity" and "insecticidal activity" are used synonymously to refer to activity of an organism or a substance (such as, for example, a protein) that can be measured by numerous parameters including, but not limited to, pest mortality, pest

weight loss, pest attraction, pest repellency, and other behavioral and physical changes of a pest after feeding on and/or exposure to the organism or substance for an appropriate length of time. For example, "pesticidal proteins" are proteins that display pesticidal activity by themselves or in combination with other proteins.

(11) As used herein, "insert DNA" refers to the heterologous DNA within the expression cassettes used to transform the plant material while "flanking DNA" can exist of either genomic DNA naturally present in an organism such as a plant, or foreign (heterologous) DNA introduced via the transformation process which is extraneous to the original insert DNA molecule, e.g. fragments associated with the transformation event. A "flanking region" or "flanking sequence" as used herein refers to a sequence of at least 20 bp, for some embodiments, at least 50 bp, and up to 5000 bp, which is located either immediately upstream of and contiguous with or immediately downstream of and contiguous with the original foreign insert DNA molecule.

(12) In an embodiment, the junction sequences of Event DP-202216-6, for example, may include polymorphisms (e.g., SNPs) or mutations that may occur spontaneously in the endogenous genomic region of the junction sequence. These may include insertion, deletion or substitution of one or more nucleotides in the junction sequence. Polynucleotide sequences that are at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, and 99% to one or more of the junction sequences.

(13) As used herein, "heterologous" in reference to a nucleic acid sequence is a nucleic acid sequence that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a heterologous nucleotide sequence can be from a species different from that from which the nucleotide sequence was derived, or, if from the same species, the promoter is not naturally found operably linked to the nucleotide sequence. A heterologous protein may originate from a foreign species, or, if from the same species, is substantially modified from its original form by deliberate human intervention.

(14) The term "regulatory element" refers to a nucleic acid molecule having gene regulatory activity, i.e. one that has the ability to affect the transcriptional and/or translational expression pattern of an operably linked transcribable polynucleotide. The term "gene regulatory activity" thus refers to the ability to affect the expression of an operably linked transcribable polynucleotide molecule by affecting the transcription and/or translation of that operably linked transcribable polynucleotide molecule. Gene regulatory activity may be positive and/or negative and the effect may be characterized by its temporal, spatial, developmental, tissue, environmental, physiological, pathological, cell cycle, and/or chemically responsive qualities as well as by quantitative or qualitative indications.

(15) "Promoter" refers to a nucleotide sequence capable of controlling the expression of a coding sequence or functional RNA. In general, a coding sequence is located 3′ to a promoter sequence. The promoter sequence comprises proximal and more distal upstream elements, the latter elements are often referred to as enhancers. Accordingly, an "enhancer" is a nucleotide sequence that can stimulate promoter activity and may be an innate element of the promoter or a heterologous element inserted to enhance the level or tissue-specificity of a promoter. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic nucleotide segments. It is understood by those skilled in the art that different regulatory elements may direct the expression of a gene in different tissues or cell types, or at different stages of development, or in response to different environmental conditions. Promoters that cause a nucleic acid fragment to be expressed in most cell types at most times are commonly referred to as "constitutive promoters".

(16) A DNA construct is an assembly of DNA molecules linked together that provide one or more expression cassettes. The DNA construct may be a plasmid that is enabled for self-replication in a bacterial cell and contains various endonuclease enzyme restriction sites that are useful for introducing DNA molecules that provide functional genetic elements, i.e., promoters, introns,

leaders, coding sequences, 3′ termination regions, among others; or a DNA construct may be a linear assembly of DNA molecules, such as an expression cassette. The expression cassette contained within a DNA construct comprises the necessary genetic elements to provide transcription of a messenger RNA. The expression cassette can be designed to express in prokaryote cells or eukaryotic cells. Expression cassettes of the embodiments are designed to express in plant cells.

(17) The DNA molecules disclosed herein are provided in expression cassettes for expression in an organism of interest. The cassette will include 5′ and 3′ regulatory sequences operably linked to a coding sequence. "Operably linked" means that the nucleic acid sequences being linked are contiguous and, where necessary to join two protein coding regions, contiguous and in the same reading frame. Operably linked is intended to indicate a functional linkage between a promoter and a second sequence, wherein the promoter sequence initiates and mediates transcription of the DNA sequence corresponding to the second sequence. The cassette may additionally contain at least one additional gene to be co-transformed into the organism. Alternatively, the additional gene(s) can be provided on multiple expression cassettes or multiple DNA constructs.

(18) The expression cassette may include in the 5′ to 3′ direction of transcription: a transcriptional and translational initiation region, a coding region, and a transcriptional and translational termination region functional in the organism serving as a host. The transcriptional initiation region (i.e., the promoter) may be native or analogous, or foreign or heterologous to the host organism. Additionally, the promoter may be the natural sequence or alternatively a synthetic sequence. The expression cassettes may additionally contain 5′ leader sequences in the expression cassette construct. Such leader sequences can act to enhance translation.

(19) A transgenic "event" is produced by transformation of plant cells with a heterologous DNA construct(s), including a nucleic acid expression cassette that comprises a transgene of interest, the regeneration of a population of plants resulting from the insertion of the transgene into the genome of the plant, and selection of a particular plant characterized by insertion into a particular genome location. An event is characterized phenotypically by the expression of the transgene. At the genetic level, an event is part of the genetic makeup of a plant. The term "event" also refers to progeny produced by a sexual outcross between the transformant and another variety that include the heterologous DNA. Even after repeated back-crossing to a recurrent parent, the inserted DNA and flanking DNA from the transformed parent is present in the progeny of the cross at the same chromosomal location. The term "event" also refers to DNA from the original transformant comprising the inserted DNA and flanking sequence immediately adjacent to the inserted DNA that would be expected to be transferred to a progeny that receives inserted DNA including the transgene of interest as the result of a sexual cross of one parental line that includes the inserted DNA (e.g., the original transformant and progeny resulting from selfing) and a parental line that does not contain the inserted DNA.

(20) Corn plant containing event DP-202216-6 may be bred by first sexually crossing a first parental corn plant consisting of a corn plant grown from event DP-202216-6 corn plant and progeny thereof derived from transformation with the expression cassettes of the embodiments that increase yield when compared to a control plant, and a second parental corn plant that does not have such constructs, thereby producing a plurality of first progeny plants; and then selecting a first progeny plant that demonstrates yield increase; and selfing the first progeny plant, thereby producing a plurality of second progeny plants; and then selecting from the second progeny plants plant with yield increase.

(21) As used herein, the term "plant" includes reference to whole plants, parts of plants, plant organs (e.g., leaves, stems, roots, etc.), seeds, plant cells, and progeny of same. In some embodiments, parts of transgenic plants comprise, for example, plant cells, protoplasts, tissues, callus, embryos as well as flowers, stems, fruits, leaves, and roots originating in transgenic plants or their progeny previously transformed with a DNA molecule disclosed herein, and therefore

consisting at least in part of transgenic cells.
(22) As used herein, the term "plant cell" includes, without limitation, seeds, suspension cultures, embryos, meristematic regions, callus tissue, leaves, roots, shoots, gametophytes, sporophytes, pollen, and microspores. The class of plants that may be used is generally as broad as the class of higher plants amenable to transformation techniques, including both monocotyledonous and dicotyledonous plants.
(23) The present disclosure provides a commodity product that is derived from a corn plant comprising event DP-202216-6. As used herein, a "commodity product" generally refers to any composition or material that includes material derived or processed from a plant, seed, plant cell, or plant part comprising event DP-202216-6. Commodity products may be viable (e.g., seeds) or nonviable (e.g., corn meal). Nonviable commodity products include but are not limited to nonviable seeds and grains; processed seeds, seed parts, and plant parts; dehydrated plant tissue, frozen plant tissue, and processed plant tissue; seeds and plant parts processed for animal feed for terrestrial and/or aquatic animal's consumption, oil, meal, flour, flakes, bran, fiber, milk, cheese, paper, cream, wine, ethanol, and any other food for human consumption; and biomasses and fuel products. Viable commodity products include but are not limited to seeds and plant cells.
(24) "Transformation" refers to the transfer of a nucleic acid fragment into the genome of a host organism, resulting in genetically stable inheritance. Host organisms containing the transformed nucleic acid fragments are referred to as "transgenic" organisms. Examples of methods of plant transformation include *Agrobacterium*-mediated transformation (De Blaere et al. (1987) *Meth. Enzymol.* 143:277) and particle-accelerated or "gene gun" transformation technology (Klein et al. (1987) *Nature* (London) 327:70-73; U.S. Pat. No. 4,945,050, incorporated herein by reference). Additional transformation methods are disclosed below.
(25) As used herein, the term "progeny" in the context of DP-202216-6 denotes the offspring of any generation of a parent plant which comprises corn event DP-202216-6.
(26) Isolated polynucleotides disclosed herein may be incorporated into recombinant constructs, typically DNA constructs, which are capable of introduction into and replication in a host cell. Such a construct may be a vector that includes a replication system and sequences that are capable of transcription and translation of a polypeptide-encoding sequence in a given host cell. A number of vectors suitable for stable transfection of plant cells or for the establishment of transgenic plants have been described in, e.g., Pouwels et al., (1985; Supp. 1987) *Cloning Vectors: A Laboratory Manual*, Weissbach and Weissbach (1989) *Methods for Plant Molecular Biology*, (Academic Press, New York); and Flevin et al., (1990) *Plant Molecular Biology Manual*, (Kluwer Academic Publishers). Typically, plant expression vectors include, for example, one or more cloned plant genes under the transcriptional control of 5′ and 3′ regulatory sequences and a dominant selectable marker. Such plant expression vectors also can contain a promoter regulatory region (e.g., a regulatory region controlling inducible or constitutive, environmentally- or developmentally-regulated, or cell- or tissue-specific expression), a transcription initiation start site, a ribosome binding site, an RNA processing signal, a transcription termination site, and/or a polyadenylation signal.
(27) During the process of introducing an insert into the genome of plant cells, it is not uncommon for some deletions or other alterations of the insert and/or genomic flanking sequences to occur. Thus, the relevant segment of the plasmid sequence provided herein might comprise some minor variations. The same is true for the flanking sequences provided herein. Thus, a plant comprising a polynucleotide having some range of identity with the subject flanking and/or insert sequences is within the scope of the subject disclosure. Identity to the sequence of the present disclosure may be a polynucleotide sequence having at least 65% sequence identity, for some embodiments at least 70% sequence identity, for some embodiments at least 75% sequence identity, for some embodiments at least 80% identity, and for some embodiments at least 85% 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% sequence identity with a sequence

exemplified or described herein. Hybridization and hybridization conditions as provided herein can also be used to define such plants and polynucleotide sequences of the subject disclosure. The sequence which comprises the flanking sequences plus the full insert sequence can be confirmed with reference to the deposited seed.

(28) A “probe” is an isolated nucleic acid to which is attached a conventional detectable label or reporter molecule, e.g., a radioactive isotope, ligand, chemiluminescent agent, or enzyme. Such a probe is complementary to a strand of a target nucleic acid, for example, to a strand of isolated DNA from corn event DP-202216-6 whether from a corn plant or from a sample that includes DNA from the event. Probes may include not only deoxyribonucleic or ribonucleic acids but also polyamides and other probe materials that bind specifically to a target DNA sequence and can be used to detect the presence of that target DNA sequence.

(29) “Primers” are isolated nucleic acids that anneal to a complementary target DNA strand by nucleic acid hybridization to form a hybrid between the primer and the target DNA strand, then extended along the target DNA strand by a polymerase, e.g., a DNA polymerase. Primer pairs refer to their use for amplification of a target nucleic acid sequence, e.g., by PCR or other conventional nucleic-acid amplification methods. “PCR” or “polymerase chain reaction” is a technique used for the amplification of specific DNA segments (see, U.S. Pat. Nos. 4,683,195 and 4,800,159; herein incorporated by reference).

(30) A nucleic acid molecule is said to be the “complement” of another nucleic acid molecule if they exhibit complete complementarity or minimal complementarity. As used herein, molecules are said to exhibit “complete complementarity” when every nucleotide of one of the molecules is complementary to a nucleotide of the other. Two molecules are said to be “minimally complementary” if they can hybridize to one another with sufficient stability to permit them to remain annealed to one another under at least conventional “low-stringency” conditions. Similarly, the molecules are said to be “complementary” if they can hybridize to one another with sufficient stability to permit them to remain annealed to one another under conventional “high-stringency” conditions. Conventional stringency conditions are described by Sambrook et al., 1989, and by Haymes et al., In: *Nucleic Acid Hybridization, a Practical Approach*, IRL Press, Washington, D.C. (1985), departures from complete complementarity are therefore permissible, as long as such departures do not completely preclude the capacity of the molecules to form a double-stranded structure. In order for a nucleic acid molecule to serve as a primer or probe it needs to be sufficiently complementary in sequence to be able to form a stable double-stranded structure under the particular solvent and salt concentrations employed.

(31) In hybridization reactions, specificity is typically the function of post-hybridization washes, the relevant factors being the ionic strength and temperature of the final wash solution. The thermal melting point (T.sub.m) is the temperature (under defined ionic strength and pH) at which 50% of a complementary target sequence hybridizes to a perfectly matched probe. For DNA-DNA hybrids, the T.sub.m can be approximated from the equation of Meinkoth and Wahl (1984) *Anal. Biochem.* 138:267-284: T.sub.m=81.5° C.+16.6 (log M)+0.41 (% GC)−0.61 (% form)−500/L; where M is the molarity of monovalent cations, % GC is the percentage of guanosine and cytosine nucleotides in the DNA, % form is the percentage of formamide in the hybridization solution, and L is the length of the hybrid in base pairs. T.sub.m is reduced by about 1° C. for each 1% of mismatching; thus, T.sub.m, hybridization, and/or wash conditions can be adjusted to hybridize to sequences of the desired identity. For example, if sequences with >90% identity are sought, the T.sub.m can be decreased 10° C. Generally, stringent conditions are selected to be about 5° C. lower than the T.sub.m for the specific sequence and its complement at a defined ionic strength and pH. However, severely stringent conditions can utilize a hybridization and/or wash at 1, 2, 3, or 4° C. lower than the T.sub.m; moderately stringent conditions can utilize a hybridization and/or wash at 6, 7, 8, 9, or 10° C. lower than the T.sub.m; low stringency conditions can utilize a hybridization and/or wash at 11, 12, 13, 14, 15, or 20° C. lower than the T.sub.m.

(32) Using the equation, hybridization and wash compositions, and desired T.sub.m, those of ordinary skill will understand that variations in the stringency of hybridization and/or wash solutions are inherently described. If the desired degree of mismatching results in a T.sub.m of less than 45° C. (aqueous solution) or 32° C. (formamide solution), it is preferred to increase the SSC concentration so that a higher temperature can be used. An extensive guide to the hybridization of nucleic acids is found in Tijssen (1993) *Laboratory Techniques in Biochemistry and Molecular Biology Hybridization with Nucleic Acid Probes*, Part I, Chapter 2 (Elsevier, New York); and Ausubel et al., eds. (1995) and Sambrook et al. (1989).
(33) The principle of hybridization analysis is that a single-stranded DNA or RNA molecule of a known sequence (e.g., the probe) can base-pair to a second DNA or RNA molecule that contains a complementary sequence (the target), with the stability of the hybridization depending on the extent of base pairing that occurs under the conditions tested. Appropriate stringency conditions for DNA hybridization, include for example, 6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by a wash of 2.0×SSC at 50° C., are known to those skilled in the art or can be found in Current Protocols in Molecular Biology, John Wiley & Sons, N.Y. (1989), 6.3.1-6.3.6. For example, the salt concentration in the wash step can be selected from a low stringency of about 2.0×SSC at 50° C. to a high stringency of about 0.2×SSC at 50° C. or up to 0.1×SSC or 0.2×SSC, at 55° C. or 65° C. In addition, the temperature in the wash step can be increased from low stringency conditions at room temperature, about 22° C., to high stringency conditions at about 65° C. Both temperature and salt may be varied, or either the temperature or the salt concentration may be held constant while the other variable (e.g., time) is changed. In one embodiment, a nucleic acid of the present disclosure will specifically hybridize to one or more of the nucleic acid molecules or complements or fragments thereof under high stringency conditions. The hybridization of the probe to the target DNA molecule can be detected by methods known to those skilled in the art. These can include, but are not limited to, fluorescent tags, radioactive tags, antibody-based tags, and chemiluminescent tags.
(34) In some embodiments, a complementary sequence has the same length as the nucleic acid molecule to which it hybridizes. In some embodiments, the complementary sequence is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotides longer or shorter than the nucleic acid molecule to which it hybridizes. In some embodiments, the complementary sequence is 1%, 2%, 3%, 4%, or 5% longer or shorter than the nucleic acid molecule to which it hybridizes. In some embodiments, a complementary sequence is complementary on a nucleotide-for-nucleotide basis, meaning that there are no mismatched nucleotides (each A pairs with a T and each G pairs with a C). In some embodiments, a complementary sequence comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or less mismatches. In some embodiments, the complementary sequence comprises 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, or 10% or less mismatches.
(35) “Percent (%) sequence identity” with respect to a reference sequence (subject) is determined as the percentage of amino acid residues or nucleotides in a candidate sequence (query) that are identical with the respective amino acid residues or nucleotides in the reference sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any amino acid conservative substitutions as part of the sequence identity. Alignment for purposes of determining percent sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2. Those skilled in the art can determine appropriate parameters for aligning sequences, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (e.g., percent identity of query sequence=number of identical positions between query and subject sequences/total number of positions of query sequence×100). For example, Clustal W method of aligning multiple sequences is described in Thompson J, Higgins D and Gibson T (1994). Clustal

W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting.” Nucleic Acids Research, Vol 22: pp. 4673-80. Another method is Clustal V, described in Higgins D G and Sharp P M (1989). “Fast and sensitive multiple sequence alignments on a microcomputer.” CABIOS, Vol. 5, No. 2: pp. 151-153.

(36) Regarding the amplification of a target nucleic acid sequence (e.g., by PCR) using a particular amplification primer pair, stringent conditions permit the primer pair to hybridize only to the target nucleic-acid sequence to which a primer having the corresponding wild-type sequence (or its complement) would bind to produce a unique amplification product, the amplicon, in a DNA thermal amplification reaction.

(37) The term “allele” refers to an alternative form of a gene, whereby two genes can differ in DNA sequences. Such differences may result from at least one mutation (e.g., deletion, insertion, and/or substitution) in the nucleic acid sequence. Alleles may result in modified mRNAs or polypeptides whose structure or function may or may not be modified. Any given gene may have none, one, or many allelic forms. Each of these types of changes may occur alone, or in combination with the others, one or more times in a given sequence.

(38) A hybridization reaction using a probe specific to a sequence found within the amplicon is yet another method used to detect the amplicon produced by a PCR reaction. The term “zygosity” generally refers to the similarity of alleles for a gene or trait in an organism (e.g., a plant). If both alleles are the same, the organism is homozygous for the allele. If the two alleles are different, the organism is heterozygous for the gene or trait. If one allele is not present, the organism is hemizygous. If both alleles are not present, the organism is nullizygous. For example, a plant is homozygous for the trait of interest if the insert DNA along with the junction sequence is present at the same location on each chromosome of a chromosome pair (both the alleles). For example, a maize plant having Event DP-202216-6 at the same location on both the copies of the chromosome. Similarly, a plant is considered heterozygous if the transgene insert along with the junction sequence (e.g., Event DP-202216-6) is present on only one of the chromosomes of a chromosome pair (only one allele). A wild-type plant is considered “null” when compared to the transgenic Event DNA.

(39) The term “label” when used herein refers to a detectable compound or composition that is conjugated directly or indirectly to a probe to generate a “labeled” probe. The label may be detectable by itself (e.g. radioisotope labels or fluorescent labels) or, in the case of an enzymatic label, may catalyze chemical alteration of a substrate compound or composition that is detectable (e.g., avidin-biotin).

(40) As used herein, a “line” is a group of plants that display little or no genetic variation between individuals for at least one trait. Such lines may be created by several generations of self-pollination and selection, or vegetative propagation from a single parent using tissue or cell culture techniques.

(41) As used herein, the terms “cultivar” and “variety” are synonymous and refer to a line which is used for commercial production. “Stability” or “stable” means that with respect to the given component, the component is maintained from generation to generation and, for some embodiments, at least three generations at substantially the same level, e.g., for some embodiments ±15%, for some embodiments ±10%, most for some embodiments ±5%. The stability may be affected by temperature, location, stress and the time of planting.

(42) “Agronomically elite” means that a line has desirable agronomic characteristics such as maturity, disease resistance, standability, ear height, plant height, and the like, in addition to yield increase due to the subject event(s).

(43) In some embodiments the DP-202216-6 maize event may further comprise a stack of additional traits. Plants comprising stacks of polynucleotide sequences can be obtained by either or both of traditional breeding methods or through genetic engineering methods. These methods include, but are not limited to, breeding individual lines each comprising a polynucleotide of

interest, transforming a transgenic plant comprising a gene disclosed herein with a subsequent gene and co-transformation of genes into a single plant cell. As used herein, the term "stacked" includes having the multiple traits present in the same plant (i.e., both traits are incorporated into the nuclear genome, one trait is incorporated into the nuclear genome and one trait is incorporated into the genome of a plastid or both traits are incorporated into the genome of a plastid). Additional traits can include for example, drought tolerance and other abiotic stress tolerance traits. Such traits can be introduced by breeding with maize plants containing other recombinant events or with maize plants containing native variations or genome edited variations.

(44) In some embodiments, DP-202216-6 maize event can be stacked with one or more additional input traits (e.g., herbicide resistance, fungal resistance, virus resistance, stress tolerance, disease resistance, male sterility, stalk strength, and the like) or output traits (e.g., increased yield, modified starches, improved oil profile, balanced amino acids, high lysine or methionine, increased digestibility, improved fiber quality, drought resistance, and the like). In a further embodiment, the DP-202216-6 maize event may be combined with one or more additional Bt insecticidal toxins or other non-Bt insecticidal proteins.

(45) In an aspect, a corn field has plants that have the MAD S-box transcription factor disclosed herein along with introduced genetic modifications affecting stature, and includes a planting density of at least 10,000 corn plants per acre. In another aspect, a corn field comprises a planting density of at least 30,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 32,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 34,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 36,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 38,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 40,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 42,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 44,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 46,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 48,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 50,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 52,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 54,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 56,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 58,000 corn plants per acre. In another aspect, a corn field includes a planting density of at least 60,000 corn plants per acre.

(46) In an aspect, a corn field has plants that have the MADS-box transcription factor, the corn plants containing MADS-box transcription factor expressing at a higher level in combination with shorter stature or semi-dwarf phenotype exhibits plant that is at least about 10%, between 10% and 15%, at least about 15%, between 15% and 20%, at least about 20%, between 20% and 25%, at least about 25%, between 25% and 30%, at least about 30%, between 30% and 35%, at least about 35%, between 35% and 40%, at least about 40%, between 40% and 45%, at least about 45%, between 45% and 50%, at least about 50%, between 50% and 55%, at least about 55%, between 55% and 60%, at least about 60%, between 60% and 65%, at least about 65%, between 65% and 70%, at least about 70%, between 70% and 75%, at least about 75%, between 75% and 80%, when compared to the plant height of control corn plants not comprising the stature genetic modification, when measured at or near reproductive/late reproductive growth phase (e.g., R1-R6).

(47) In an aspect, a corn field has plants that have the MADS-box transcription factor disclosed herein along with introduced genetic modifications affecting stature and includes a planting density of between 10,000 and 50,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 10,000 and 40,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 10,000 and 30,000 corn plants per acre. In an aspect, a corn field

includes a planting density of between 10,000 and 25,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 10,000 and 20,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 60,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 58,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 55,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 50,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 45,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 42,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 40,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 38,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 36,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 34,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 32,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 20,000 corn plants and 30,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 24,000 corn plants and 58,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 38,000 corn plants and 60,000 corn plants per acre. In an aspect, a corn field includes a planting density of between 38,000 corn plants and 50,000 corn plants per acre.

(48) In some embodiments, corn plants containing DP-202216-6 event can be crossed with corn plants containing other corn Events or combination thereof and the resulting properties of the progeny plants are evaluated. For example, corn plants containing DP-202216-6 Event can be crossed or combined with corn plants including one or more combinations, of the following: MON810; DAS-59122-7; MIR604; MON89034; MON863; MON87411; MON87403; MON87427; MON-00603-6 (NK603); MON-87460-4; MON-88017-3; LY038; TC1507; 5307; DAS-06275-8; BT176; BT11; MIR162; GA21; MZDT09Y; SYN-05307-1; DP-004114-3; and DAS-40278-9.

(49) The following examples are offered by way of illustration and not by way of limitation. As described herein, Event DP-202216-6 is also referred to as “Event 16”, “E16” “event 16” or “Event 16-6” and they all refer to the same maize event DP-202216-6. The protein encoded by the Maize MADS box ZmM28 gene in the plasmid PHP40099 or the Event DP-202216-6 is also referred to as AG099 protein and the corresponding DNA sequence as AG099 gene or AG099 DNA.

EXAMPLES

Example 1

Field Performance of Reduced Stature Maize Plants Having Increased and Extended Expression of AG099

(50) A series of grain yield trials are conducted in elite corn hybrids across testing locations in order to assess the yield in elite reduced stature maize hybrids and reduced stature maize hybrids having AG099 events. For example, hybrids with maturities ranging from 105-112 days are used to evaluate the performance of reduced stature AG099 events relative to controls. In certain aspects, locations selected to achieve various yield levels can range from highly drought stressed 70 bu/acre to optimal growing conditions 250 bu/acre. Soil types include a variety of high sand, sandy loam, silty loam, loam and some clay. In such locations, entries containing the AG099 construct are compared to its control (e.g., AG099 alone or reduced stature plants alone, but not the combination) for each specific hybrid background. Two to four replicates of a split plot design are established at each location. A mixed model analysis of variance can be conducted using ASREML where BLUEs (Best Linear Unbiased Estimates) are generated for each AG099 event in combination with reduced stature or height lines and respective controls across hybrids within a given location. Pairwise contrasts of these event BLUEs to wild type BLUEs are analyzed to test significant differences.

(51) In certain aspects, such as for example, high-yielding where yields of over 160 bu/acre are considered as normal and often can be greater than 180 bu/acre which represents a large portion of the most productive corn growing regions in the United States. In order to evaluate the response of reduced stature corn hybrids containing AG099 event or construct to drought stress, additional sites in Kansas, Texas and California can also be established with the capability of specifically managing the amount of water applied to the test plots during the growing season. Managed stress conditions can range from severe stress of less than 120 bu/acre yield up to a very mild stress where the yield is under 160 bu/acre.

Example 2

Field Performance of Reduced Stature Maize Plants Having Increased and Extended Expression of AG099 Under Varying Planting Densities and Row Spacing

(52) To test AG099 containing events with reduced stature and resulting yield increase under different planting populations, experiments are conducted at multiple locations and at varying planting population or planting density. Experimental treatments include planting populations of for example, 36,000, 40,000, 44,000, 48,000, 52,000, 56,000, 60,000, 70,000 or 75,000 plants per acre. Within each population, in certain experiments corn Event DP-202216-6 bred for reduced stature and the corresponding control (e.g., Event DP-202216-6 alone or the reduced stature line without the Event) are evaluated across multiple hybrid backgrounds. The grain yield of reduced stature corn having Event DP-202216-6 per acre within and across densities are measured by calculating the difference in yield (BLUEs) of the test line to that of the control.

(53) To test AG099 containing events with reduced stature and resulting yield increase under different row width, experiments are conducted at multiple locations and at varying row width. Experimental seeding rates include planting densities of e.g., 36,000, 40,000, 44,000, 48,000, 52,000, 56,000, 60,000, 64,000, 65,000, 70,000 or 75,000 plants per acre at varying row width or row spacing. For example, row width can range from about 8 inches, 10 inches, 12 inches, 14 inches, 16 inches, 18 inches, 20 inches, 22 inches, 24 inches, 26 inches, 28 inches, 30 inches or other row widths that are available in mechanical planting equipment (seed planters). In addition, a twin row configuration, where two rows are e.g., 7-8″ apart are also possible. Within each population, in certain experiments corn Event DP-202216-6 bred for reduced stature and the corresponding control (e.g., Event DP-202216-6 alone or the reduced stature line without the Event) are evaluated across multiple hybrid backgrounds. The grain yield of reduced stature corn having Event DP-202216-6 per acre within and across densities are measured by calculating the difference in yield (BLUEs) of the test line to that of the control.

(54) TABLE-US-00002 TABLE 2 Predicted Field Yield and Plant Height for Short Stature Plants - Multilocation Field Trials Among Multiple Genetic Backgrounds % Yield % Plant Reduction Height Reduction Experimental Lines Compared to Control Compared to Control D8 short stature - 6.95 31.33 Genetic Background 1 D8 short stature - 9.56 33.98 Genetic Background 2 D8 short stature - 7.92 34.43 Genetic Background 3 D8 short stature - 8.53 30.64 Genetic Background 4 D8 short stature - 8.73 36.49 Genetic Background 5

(55) As part of the multilocation trials, in one of the locations, genetic backgrounds 4 and 5 for D8 had a statistically significant increase in yield compared to the NULL control (P<0.10 confidence).

Example 3

Secondary Trait Characteristics of Reduced Stature Maize Plants Containing Increased Expression of AG099 (e.g., Event DP-202216-6)—Inbred and Hybrid Analysis

(56) Secondary agronomic characteristics are observed for reduced stature maize plants that exhibit increased and extended expression of AG099. Agronomic parameters such as plant, ear height, early or late root lodging, brittle snap, grain moisture, anthesis-silking interval and others are measured. Growing degree units to silk (GDUSLK): Measurement records the total accumulated growing degree units when 50% of the plants in the plot have fully emerged silks. A single day equivalent is approximately 2.5 growing degrees units (GDU) for this data set. Growing degree

units to shed (GDUSHD): Measurement records the total accumulated growing degree units when 50% of the plants in the plot have tassels that are shedding pollen. A single day equivalent is approximately 2.5 growing degrees units for this data set.
(57) Ear height (EARHT): Measurement from the ground to the attachment point of the highest developed ear on the plant. Ear height is measured in inches.
(58) Plant height (PLTHT): Measurement from the ground to the base of the flag leaf. Plant height is measured in inches.
(59) Moisture (MST): Measurement of the percent grain moisture at harvest.
(60) Yield: Recorded weight of grain harvested from each plot. Calculations of reported bu/acre yields are made by adjusting to measured moisture of each plot.
(61) Agronomic data and observations are collected for the inbred trials and analyzed for comparison to a wild type entry (WT), or without the AG099 trait version of the same genotype or with respect to reduced stature, appropriate controls are used.
(62) To evaluate the hybrid data, a mixed model framework is used to perform multi location analysis. In the multi-location analysis, main effect construct design is considered as fixed effect. Factors for location, background, tester, event, background by construct design, tester by construct design, tester by event, location by background, location by construct design, location by tester, location by background by construct design, location by tester by construct design, location by event, location by tester by event are considered as random effects. The spatial effects including range and plot within locations are considered as random effects to remove the extraneous spatial noise. The heterogeneous residual is assumed with autoregressive correlation as AR1*AR1 for each location. The estimate of construct design and prediction of event for each background are generated. The T-tests are conducted to compare construct design/event with WT. A difference is considered statistically significant if the P-value of the difference is less than 0.05. Yield analysis is by ASREML (VSN International Ltd; Best Linear Unbiased Prediction; Cullis, B. R et al (1998) *Biometrics* 54: 1-18, Gilmour, A. R. et al (2009); ASRem1 User Guide 3.0, Gilmour, A. R., et al (1995) *Biometrics* 51: 1440-50).
(63) To evaluate the inbred data, a mixed model framework is used to perform multi location analysis. In the multi-location analysis, main effect construct design is considered as fixed effect. Factors for location, background, event, background by construct design, location by background, location by construct design, location by background by construct design, location by event and rep within location are considered as random effects. The spatial effects including range and plot within locations are considered as random effects to remove the extraneous spatial noise. The heterogeneous residual is assumed with autoregressive correlation as AR1*AR1 for each location. The estimate of construct design and prediction of event for each background are generated. The T-tests were conducted to compare construct design/event with WT. A difference is considered statistically significant if the P-value of the difference is less than 0.10. Yield analysis was by ASREML (VSN International Ltd; Best Linear Unbiased Prediction; Cullis, B. Ret al (1998) *Biometrics* 54: 1-18, Gilmour, A. R. et al (2009); ASRem1 User Guide 3.0, Gilmour, A. R., et al (1995) *Biometrics* 51: 1440-50).
(64) TABLE-US-00003 TABLE 3 Field Standability Measurements for Plants Containing D8 Stature Genetic Modification Among Multiple Genetic Backgrounds % Difference in Standability Compared to Control Plants Experimental Lines Loc 1 Loc 2 Loc 3 Loc 4 Loc 5 Loc 6 D8 short stature - 40.68* 19.25* −0.83 22.64* −1.84 −8.95 Genetic Background 1 D8 short stature - 35.07* −18.39 0.04 21.04* −4.57 −8.96 Genetic Background 2 D8 short stature - 38.69* 20.61* 0.69 25.38* −0.94 — Genetic Background 3 D8 short stature - 36.0* — −2.36 34.61* −1.03 — Genetic Background 4 D8 short stature - 44.07* 15.17* −1.05 15.99* −1.90 −8.96 Genetic Background 5
(65) As part of the multilocation trials, the data noted with an asterisk (*) denote statistically significant difference compared to the NULL control (P<0.10 confidence).
(66) TABLE-US-00004 TABLE 4 Agronomic measurements for D8 in combination with AG099

from Multilocation Field Trials. BLUE Difference BLUE Difference BLUE Difference Compared to D8 Compared to D8 Compared to D8 Experimental Single Control Single Control Single Control Line (Yield) (Ear Height) (Plant Height) D8 + AG099 4.86 0.52 −1.79
(67) As part of the multilocation trials, in one of the locations, D8 had a statistically significant increase in yield compared to the NULL control (P<0.10 confidence). For yield the BLUE DIFF is bu/acre and for Ear height and Plant height the units are inches.
(68) Experimental designs of the D8 experiments were split plots with hybrid background as the main plot and the D8*AG099 or D8 version of the hybrid as the sub plot. Two replicates were established at five testing locations with main plots randomized within replication and sub plots randomized within main plot. Experimental entries were grown in four-row plots that ranged from 4.4 m to 5.3 m in length with a 0.5 m alley in between. A mixed model analysis of variance was conducted using ASREML accounting for random field and spatial components as well as the fixed components. BLUEs were generated for both the D8*AG099 stacks and the D8 single entries across all hybrid backgrounds. Pairwise contrasts of the D8*AG099 and D8 single BLUEs conducted to test for significant differences (BLUE DIFFs) at P<0.10
Example 4
Nutrient Management of Reduced Stature AG099 Plants
(69) Reduced stature maize AG099 expressing plants are grown at higher seeding rates. Further, such plants may exhibit early growth vigor and increased grain fill. Appropriate nutrient management considerations include:
(70) Early and late season management practice for AG099 expressing short stature corn plants include for example, reducing starter N and increasing side-dressing N application between V6 to V12, and delaying single application of N fertilizer and splitting N fertilizer applications at either the Agronomic Optimum N Rate (AONR) or the Economic Optimum N Rates (EONR). This management practice can also result in reduced overall Nitrogen application throughout the growing season per net yield of bu/acre. Over the top N application for shorter, but high-yielding AG099 plants are also contemplated. Maize plants expressing a recombinant maize polynucleotide sequence encoding the polypeptide (SEQ ID NO: 1) are field tested in combination with a height reduction phenotype.
(71) Results can demonstrate that expression of the recombinant polynucleotide encoding SEQ ID NO: 1 in a reduced stature background can increase grain yield of maize under field conditions on a per unit of basis of applied nitrogen when compared to an appropriate control.
Example 5
Enhanced Disease & Pest Management of Reduced Stature AG099 Plants
(72) Reduced stature maize AG099 expressing plants are grown at higher seeding rates than comparable corn hybrids. The occurrence of pests and diseases increases as plant density increases. Because short stature plants are planted at higher than normal densities, likelihood of plant diseases for short stature corn is high. In addition, under high-density planting, reduced stalk strength due to the possibility of lower cellulose and lignin content may increase corn borer infestation of stalks, which results in increased stalk lodging rate. Further, higher plant density generally corresponds to reduced spatial distance between individual maize plants. This spatial crowding or denser canopy adversely affects the ventilation and light conditions of maize canopies, which in turn can increase the spread of pathogens and of stalk rot spores. Since short stature plants have a denser canopy than taller/normal plants, planting at a high density may increase disease occurrence. In addition, absorption of nutrients at high planting density, is reduced. With shorter roots, semi-swarf plants may be under a higher pressure at high planting density, thereby limiting nutrient uptake. Reduced nutrient update can in turn decrease the soluble sugar content and physiological activity of stalk, thereby reducing the resistance of stalk to rot and sheath blight.
(73) Further, such plants may exhibit early growth vigor and increased grain fill. Appropriate disease management considerations include: Seed treatment with fungicides and insecticides at a

dosage rate that is different than conventional/control corn hybrids. Early and late season, such as reducing or increasing the amount and frequency of fungicides and insecticides applications. Planting the plants earlier than a normal planting schedule to enable the plants develop into a mature state earlier and resistant to disease infection before or during exposure to fungal spores. Alternatively, prescriptive pesticide application, over-the-top of semi-dwarf AG099 plants during mid or late season.
(74) Maize plants expressing a recombinant maize polynucleotide sequence encoding the polypeptide (SEQ ID NO: 1) are field tested in combination with a height reduction phenotype, such as, for example genome edited Della (D8) variants that exhibit semi-dwarf phenotype.
Example 6
Gibberellic Acid Pathway Modification—Stature Reduction in Combination with Increased AG099 Expression
(75) Gibberellins have been identified as determinants of plant height in many plant species including maize and rice. Mutants such as sd1 in rice, rht-1 in wheat or barley sdw1 map to genes involved in gibberellin synthesis or signaling. Through CRISPR-Cas genome editing nucleic acid guided CAS endonucleases, mutations of GA pathway are introduced into more elite germplasm with minimal genetic drag associated with conventional breeding material. Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, weaker or stronger alleles of previously known mutations of GA pathway are introduced into more elite germplasm. Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, new variations of one or more components of the GA pathway are introduced into elite germplasm with minimal genetic drag associated with conventional breeding material. These targets include for example, GA1, GA3, GA4, GA7, GA20-oxidases (GA20ox), GA3-oxidases (GA3ox) and GA2-oxidases (GA2ox). Disruption of these enzymes through genome editing affects plant stature. GA20ox and GA3ox catalyze oxidations which convert inactive GAs into active GAs (GA20, GA1, GA4) and thus enhance GA responses. GA2ox deactivates GAs by converting GA4 and GA1 into inactive forms. Therefore, methods and compositions are provided that modulate the expression levels, activity levels and a combination thereof of GA these GA biosynthetic pathway that impact plant stature. More specifically, genome edited variants are provided that affect GA biosynthesis, GA signaling and/or a combination thereof.
(76) Reduction in plant height through GA pathway manipulation along with increased and extended expression of AG099 results in overall increase in yield per acre of high-density corn.
Example 7
Genome Editing of DELLA Proteins and Other GA Regulators for Stature Modification in AG099 Maize Plants
(77) DELLA proteins are a subfamily of the GRAS superfamily of proteins and play an important role in the negative regulation of GA signaling. DELLA proteins such as D8, D9, and others are suitable targets for generating variations in protein function to alter stature. Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, mutations of DELLA proteins were introduced into more elite germplasm with minimal genetic drag associated with conventional breeding material. Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, weaker or stronger alleles mutations of DELLA proteins are introduced into more elite germplasm of plants such as maize, rice, wheat, sorghum and other crop plants.
(78) TABLE-US-00005 TABLE 2 Agronomic characteristics of D8 genome edited plants. PHTYLD % % Plant % Harvest HI Hybrid (bu/acre) Red'n PHTKPE Red'n Height Red'n Index Difference Hybrid 1 226 28 528 23 145 31 0.58 0 NULL 315 681 212 0.58 (Control 1) Hybrid 2 191 35 504 28 143 34 0.58 0.03 NULL 292 697 217 0.55 (Control 2) Hybrid 3 213 33 517 27 170 29 0.54 0.01 NULL 318 704 239 0.53 (Control 3) Hybrid 4 197 37 500 29 153 36 0.54 −0.01 NULL 312 706 237 0.55 (Control 4) Across 33 27 33 0.0075 Hybrids
(79) This table shows data collected from a field trial conducted in Johnston, Iowa. This replicated

study included four elite Stiff stalk inbreds crossed to one genome edited non-stiff stalk D8 inbred, as well as their respective null comparators. In total, eight hybrids were tested. Traits that were taken at the end of the growing season and that are shown in the table are: 1) ear photometry estimated yield (PHTYLD), 2) ear photometry kernels per ear (PHTKPE), 3) Plant Height and 4) Harvest Index. For the ear photometry traits, multiple ears from each of two rows were harvested and analyzed with the standard Pioneer/Corteva ear photometry machine. Plant height was determined in the field using a measuring stick; while Harvest Index was measured as the difference in dry weight between all above ground vegetative tissue relative to the dry weight of the grain. Averaged across hybrids, the plant height of the D8 edited hybrids was reduced 33% relative to the nulls. Average estimated grain yield was also reduced in the edits by 33%, while the number of kernels per ear was decreased on average by 27%. Overall, there was minimal change in the Harvest Index.

(80) In addition to DELLAs, feedback regulators of GA biosynthesis such as for example, RSG (Repression of Shoot Growth), a bZIP transcription factor, and its interactors 14-3-3, SCL3 (Scarecrow-like3), another member of the GRAS family and those components that have been identified as GA regulators are targets for genome editing. Therefore, methods and compositions are provided that modulate the expression levels, activity levels and a combination thereof of GA regulators such as DELLA that impact plant stature. More specifically, genome edited variants are provided that affect GA regulation, GA signaling and/or a combination thereof are provided herein.

Example 8

Brassinosteroid Pathway Modification in Combination with Increased AG099 Expressing Maize Lines

(81) Brassinosteroids are a group of steroid hormones that have been identified in many plant species for a variety of functions including stature. Brassinosteroid-deficient mutants have also been a significant source of dwarfism in crops such as barley, e.g., uzu-type barley, which is insensitive to brassinosteroid treatment, has lodging resistance and upright leaf angle; *Arabidopsis* BRI1; and rice D61, encoding the brassinosteroid receptor.

(82) Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, previously known mutations of Brassinosteroid pathway are introduced into more elite germplasm with minimal genetic drag associated with conventional breeding material. Through CRISPR-cas genome editing nucleic acid guided CAS endonucleases, weaker or stronger alleles of previously known mutations of Brassinosteroid pathway are introduced into more elite germplasm of plants such as maize, rice, wheat, sorghum and other crop plants. Genome edited variants of Brassinosteroid pathway may exhibit varying degrees of one or more characteristics selected from: shortened upper internodes, shorter grain, upright leaves, delayed flowering time, delayed leaf senescence. In addition to the biosynthetic pathway, perception and signal transduction of the Brassinosteroid pathway are amenable to manipulation using the methods and compositions provided herein for modulating stature. Stronger or weaker alleles of BRI1, BRL1, BRL3 receptors are also suitable candidates for genome editing to improve stature, for example, by reducing plant height. Weaker alleles of the Brassinosteroid biosynthetic enzymes or targeting genes downstream of the major steps of the biosynthesis pathway may be helpful ways to address reducing plant height by modulating Brassinosteroid pathway, wherein the plant height reduction is not severe and semi-dwarf phenotype is obtained.

(83) Reduction in plant height through Brassinosteroid pathway manipulation along with increased and extended expression of AG099 results in overall increase in yield per acre of high-density corn.

Example 9

Weaker Alleles of Br1 Mutation and Increased AG099 Expression

(84) The candidate gene for br1 mutation mapped to c1_192.24cM with its physical location at 223,645,759-223,649,276 and there are 4 splice variants of the gene model listed in the database. Genotypic variation in 507 out of 600 analyzed maize lines (84%) at the br1 locus is covered by

four haplotypes belonging to groups 1, 2, 4, and 6.

(85) Weaker alleles of Br1 gene loci are generated by targeted mutagenesis of the Br1 genomic region and evaluating height reduction. Plants with moderate height reduction are bred with increase AG099 expressing plants.

Example 10

Mid/Late Season Application of Agriculturally Relevant Compositions Over AG099 Expressing Short Stature Field Corn Plants

(86) Higher yielding corn plants that exhibit reduced plant height are described. Specifically, AG099 corn plants (e.g., those that include heterologous expression of a MADS-box transcription factor such as Zmm28) that also exhibit shorter, dwarf, or semi-dwarf phenotype are described.

(87) Application of crop inputs such as fertilizers, nutrient, fungicides, insecticide, herbicide, or other pesticides, or other applications such as applying pollen for increased seed production, cover crop seeds during growing season of corn, or other applications to corn fields during mid- or late season are described, where such application is possible over AG099 plants exhibiting dwarf, semi-dwarf or short stature that is about 10% shorter to about 50% shorter to a comparable normal height hybrid corn plant (e.g., height of about 2 meters during the growing season).

(88) Short stature AG099 plants benefit from an extended application of crop inputs described above. For example, targeted application of herbicides to remove a particular type of weed in a late growing season can increase yield. This can be accomplished by a farm equipment that can pass through a late season corn field for over-the-top application without causing significant damage to the corn plants, especially when they are at later stages of vegetative/reproductive growth. Alternatively, an unmanned aerial vehicle, such as a drone, can also provided targeted application of a herbicide in a corn field at late season or mid-season, where the hybrid corn plants are at a height of greater than 1.5 meters, for example.

(89) Another example where short stature AG099 plants benefit from an extended application of a crop input include for example, application of late-season nutrients. Given the high-yielding attribute of AG099 plants, shorter stature of AG099 plants provide an opportunity to apply late season fertilizer to replenish nitrogen resources, for example, during the reproductive phase.

(90) Another example where short stature AG099 plants benefit from an extended application of a crop input include for example, application of mid/late-season fungicides. Mid/late-season diseases are better managed through a targeted application of one or more fungicides to minimize disease occurrence and/or disease symptoms.

Example 11

Root Growth Enhancement of AG099 Expressing Field Corn Plants that Also Exhibit Shorter Stature

(91) Shorter corn plants may also have shorter roots, thereby limiting their ability to grow deeper and acquire nutrients and water, among other components. While the reduced plant height may be advantageous for root lodging, for example due to environmental factors such as strong winds, planting density, the shorter roots may impact standability in certain soil types or soil conditions. Root lodging can occur during late vegetative stages and during later reproductive stages. In addition, root damage due to corn root worm pressure, drought conditions, soil compaction, fields where soil nitrogen has been lost due to wet condition and colder soil temperature during early season increase the likelihood of root lodging because the corn plants could not establish sufficient rooting.

(92) To enhance root growth for corn plants having both a high yielding trait such as AG099 and a short stature phenotype, one or more agronomic practices are followed. For example, a root stimulant at planting or during the early season helps enhance root growth. Such a root stimulant can be seed applied (e.g., a seed treatment biological) or soil applied. In certain embodiments, additional genetic traits including QTLs, or transgenes, or genome edited variants that increase root growth are contemplated for combinations with short stature plants.

(93) When corn plants exhibit poor standability (i.e., ability of a corn plant to stand upright to be harvested by standard farm equipment, such as a combine), harvestable yield may be lower than a field where standability is not significant. Dwarf corn plants, semi-dwarf corn plants, and or short stature corn plants, are better at resisting lodging. However, to further improve the performance and or to reduce higher % lodging or increase % standability, root growth enhancing agricultural compositions are provided for of high yielding short stature plants. Generally, “lodging” can refer to either stalk lodging or root lodging. Likelihood of stalk lodging for short stature corn plants is low.

Example 12

Hybrid Corn Seed Production Involving Female Inbred Corn Plants Exhibiting Shorter Stature and Higher Yield Due to the Heterologous Expression of MADS-Box Transcription Factor

(94) Higher yielding corn plants that are female inbred and that also exhibit reduced plant height are described. Specifically, AG099 corn plants (e.g., those that include heterologous expression of a MADS-box transcription factor such as Zmm28) that also exhibit shorter, dwarf, or semi-dwarf phenotype are described in a hybrid production setting. Seed production yield goes higher when the female parental inbred line in hybrid seed production yield higher. Female inbred lines containing an increased expression of a MADS box transcription factor that contributes to higher grain yield were also developed to exhibit shorter stature, for example up to about 10% to about 50% reduction in height compared to a typical or control female inbred corn plant.

(95) In a seed production field, AG099 female inbred plants that exhibit a reduced plant height about 20% are planted in one row. In the adjacent row, male inbred plants exhibiting normal/typical plant height for a male inbred line are planted. The seed yield is determined by measuring the seed yield of the female plants in the field and normalized for an acre. Depending the environmental and growth conditions, higher corn seed yield is expected when the female inbred parental line is shorter relative to the male inbred parental line. Reduction in plant height for AG099 expressing plants enable improved pollen flow, thereby increasing fertilization, kernel set and seed yield. Overall seed yield in a seed production field where AG099 containing female inbred lines that are shorter compared to the male inbreds, is expected to be higher than a comparable field where the AG099 containing plants are of normal height and not shorter than the male inbreds.

Claims

1. A corn plant comprising a stature reducing genetic modification and a yield enhancing genetic modification, wherein the yield enhancing genetic modification comprises a heterologous regulatory polynucleotide operably linked to a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 99% identical to SEQ ID NO: 1 and wherein the corn plant exhibits a plant height reduction of about 5% to about 50% due to the introduced genetic modification at a genomic locus encoding a D8 polypeptide represented by SEQ ID NO: 6, when compared to a control corn plant not comprising the stature reducing genetic modification.
2. The corn plant of claim 1, wherein the plant comprises event DP-202216-6, wherein a representative sample of seed of said corn event has been deposited with American Type Culture Collection (ATCC) with Accession No. PTA-124653.
3. A seed produced from the corn plant of claim 1, wherein a progeny from the seed exhibits increased yield and reduced stature when compared to the control plant.
4. A method of increasing planting density of corn plants comprising a heterologous regulatory polynucleotide operably linked to a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 99% identical to SEQ ID NO: 1, the method comprising: a) providing corn plants wherein the expression of the polynucleotide encoding the polypeptide is increased when compared to a control plant; b) reducing plant height by introducing a genetic modification at a genomic locus encoding a D8 polypeptide represented by SEQ ID NO: 6 that results in reduced

stature of the corn plants; and c) planting the corn plants at a planting density of about 30,000 to about 75,000 plants per acre.
5. The method of claim 4, wherein the corn plants comprise event DP-202216-6.
6. The method of claim 4, wherein the corn plants are planted in a plurality of rows having a row width of about 8 inches to about 30 inches.
7. The method of claim 4, wherein the corn plants yield an average of about 3 bu/acre compared to control corn plants.
8. A method of increasing nitrogen use efficiency of a population of corn plants per unit of applied nitrogen, the method comprising: a) providing the population of corn plants, wherein the expression of a polynucleotide that encodes a polypeptide comprising an amino acid sequence that is at least 99% identical to SEQ ID NO: 1 is increased due to a genetic modification compared to a control plant; b) modifying the plant height of the population of the corn plants by introducing a genetic modification at a genomic locus encoding a D8 polypeptide represented by SEQ ID NO: 6 that results in reduced stature of the population of the corn plants; and c) growing the population of plants in a crop growing environment where the applied nitrogen is about 10% to about 50% less than an application rate of about 100 lbs to about 400 lbs of nitrogen per acre.
9. The method of claim 8, wherein the corn plants comprise event DP-202216-6.
10. The method of claim 8, wherein the corn plants exhibit an increase in grain yield of an average of at least 3 bu/acre when compared to the control plant.
11. The method of claim 8, wherein the population of corn plants exhibit an increase in nitrogen use efficiency of at least about 5% when compared to the control plant.
12. The method of claim 8, wherein the nitrogen use efficiency is increased by applying late season nitrogen, wherein at least about 25% of total nitrogen applied is applied on or after V8-V12 stage.